Importing Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
plt.style.use('seaborn-v0_8-pastel')

import seaborn as sns


%matplotlib inline
# used to display your plots within the notebook.
```

Load and Explore the Dataset

```
# Importing the dataset
df = pd.read_csv(r"C:\Users\yasha\OneDrive\Desktop\inlighntech.com\My projects\Python\
New Year Sales Data.csv", encoding='latin-1')

# Checking the dataset
df.head()

   User_ID  Cust_name Product_ID Gender Age Group  Age  Marital_Status  \
0  1002903  Sanskriti  P00125942      F     26-35   28               0
1  1000732     Kartik  P00110942      F     26-35   35               1
2  1001990      Bindu  P00118542      F     26-35   35               1
3  1001425     Sudevi  P00237842      M      0-17   16               0
4  1000588       Joni  P00057942      M     26-35   28               1

            State      Zone       Occupation Product_Category  Orders  \
0     Maharashtra   Western       Healthcare             Auto       1
1  Andhra Pradesh  Southern             Govt             Auto       3
2   Uttar Pradesh   Central       Automobile             Auto       3
3       Karnataka  Southern     Construction             Auto       2
4         Gujarat   Western  Food Processing             Auto       2

    Amount  Status  unnamed1
0  23952.0     NaN       NaN
1  23934.0     NaN       NaN
2  23924.0     NaN       NaN
3  23912.0     NaN       NaN
4  23877.0     NaN       NaN

# Checking the rows and columns
df.shape

(11251, 15)

# Checking info about column names, non-null values and Data types
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
```

```
 0   User_ID           11251 non-null  int64
 1   Cust_name         11251 non-null  object
 2   Product_ID        11251 non-null  object
 3   Gender            11251 non-null  object
 4   Age Group         11251 non-null  object
 5   Age               11251 non-null  int64
 6   Marital_Status    11251 non-null  int64
 7   State             11251 non-null  object
 8   Zone              11251 non-null  object
 9   Occupation        11251 non-null  object
 10  Product_Category  11251 non-null  object
 11  Orders            11251 non-null  int64
 12  Amount            11239 non-null  float64
 13  Status            0 non-null      float64
 14  unnamed1          0 non-null      float64
dtypes: float64(3), int64(4), object(8)
memory usage: 1.3+ MB
```

Data Cleaning

```python
# Checking null values
df.isnull().sum()
```

```
User_ID                 0
Cust_name               0
Product_ID              0
Gender                  0
Age Group               0
Age                     0
Marital_Status          0
State                   0
Zone                    0
Occupation              0
Product_Category        0
Orders                  0
Amount                 12
Status              11251
unnamed1            11251
dtype: int64
```

```python
# Dropping the null values and unnecessary columns
df.drop(["Status", "unnamed1"], axis= 1, inplace=True, errors='ignore')

# Dropping rows with null values in other important columns.
important_cols = ['Gender', 'Age Group', 'Marital_Status', 'State', 'Zone',
'Occupation', 'Product_Category', 'Orders', 'Amount']

df = df.dropna(subset=important_cols)

df.isnull().sum()
```

```
User_ID             0
Cust_name           0
Product_ID          0
Gender              0
Age Group           0
Age                 0
```

```
Marital_Status      0
State               0
Zone                0
Occupation          0
Product_Category    0
Orders              0
Amount              0
dtype: int64

#Converting the Amount column to an integer type
df['Amount'] = df['Amount'].astype(int)
```

Data Overview and Summary

```
# Statistical Summary
df.describe()

            User_ID           Age  Marital_Status        Orders        Amount
count  1.123900e+04  11239.000000    11239.000000  11239.000000  11239.000000
mean   1.003004e+06     35.410357        0.420055      2.489634   9453.610553
std    1.716039e+03     12.753866        0.493589      1.114967   5222.355168
min    1.000001e+06     12.000000        0.000000      1.000000    188.000000
25%    1.001492e+06     27.000000        0.000000      2.000000   5443.000000
50%    1.003064e+06     33.000000        0.000000      2.000000   8109.000000
75%    1.004426e+06     43.000000        1.000000      3.000000  12675.000000
max    1.006040e+06     92.000000        1.000000      4.000000  23952.000000

# Checking for unique values
df.nunique()

User_ID             3752
Cust_name           1250
Product_ID          2350
Gender                 2
Age Group              7
Age                   81
Marital_Status         2
State                 16
Zone                   5
Occupation            15
Product_Category      18
Orders                 4
Amount              6583
dtype: int64

for col in ['Gender', 'Age Group', 'Marital_Status', 'State', 'Zone', 'Occupation',
'Product_Category']:
    if col in df.columns:
        print(f"\nUnique values in {col}:")
        print(df[col].unique())


Unique values in Gender:
['F' 'M']

Unique values in Age Group:
['26-35' '0-17' '18-25' '51-55' '46-50' '55+' '36-45']
```

```
Unique values in Marital_Status:
[0 1]

Unique values in State:
['Maharashtra' 'Andhra\xa0Pradesh' 'Uttar Pradesh' 'Karnataka' 'Gujarat'
 'Himachal Pradesh' 'Delhi' 'Jharkhand' 'Kerala' 'Haryana'
 'Madhya Pradesh' 'Bihar' 'Rajasthan' 'Uttarakhand' 'Telangana' 'Punjab']

Unique values in Zone:
['Western' 'Southern' 'Central' 'Northern' 'Eastern']

Unique values in Occupation:
['Healthcare' 'Govt' 'Automobile' 'Construction' 'Food Processing'
 'Lawyer' 'Media' 'Banking' 'Retail' 'IT Sector' 'Aviation' 'Hospitality'
 'Agriculture' 'Textile' 'Chemical']

Unique values in Product_Category:
['Auto' 'Hand & Power Tools' 'Stationery' 'Tupperware' 'Footwear & Shoes'
 'Furniture' 'Food' 'Games & Toys' 'Sports Products' 'Books'
 'Electronics & Gadgets' 'Decor' 'Clothing & Apparel' 'Beauty'
 'Household items' 'Pet Care' 'Veterinary' 'Office']
```

```
df = df.apply(lambda col: col.str.replace('\xa0', ' ', regex=False) if col.dtype ==
'object' else col)
```

```
print(df['State'].unique())
```

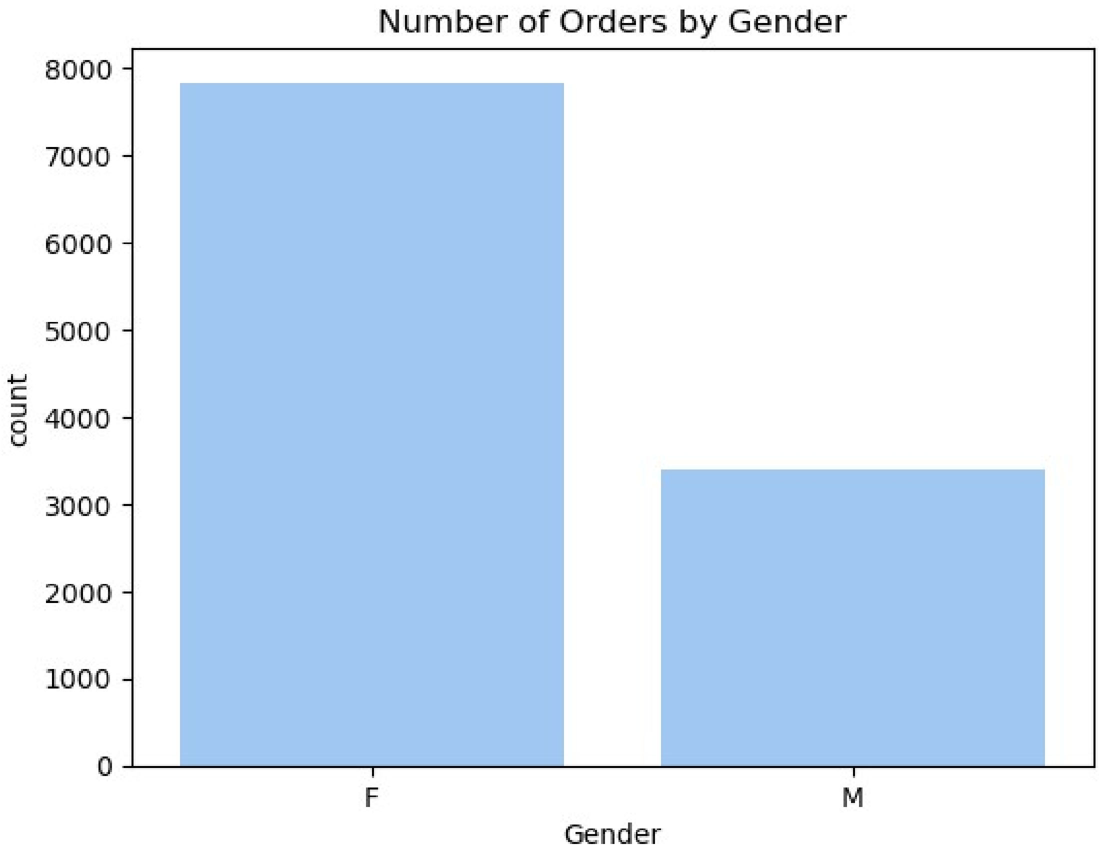
```
['Maharashtra' 'Andhra Pradesh' 'Uttar Pradesh' 'Karnataka' 'Gujarat'
 'Himachal Pradesh' 'Delhi' 'Jharkhand' 'Kerala' 'Haryana'
 'Madhya Pradesh' 'Bihar' 'Rajasthan' 'Uttarakhand' 'Telangana' 'Punjab']
```

Exploratory Data Analysis (EDA)

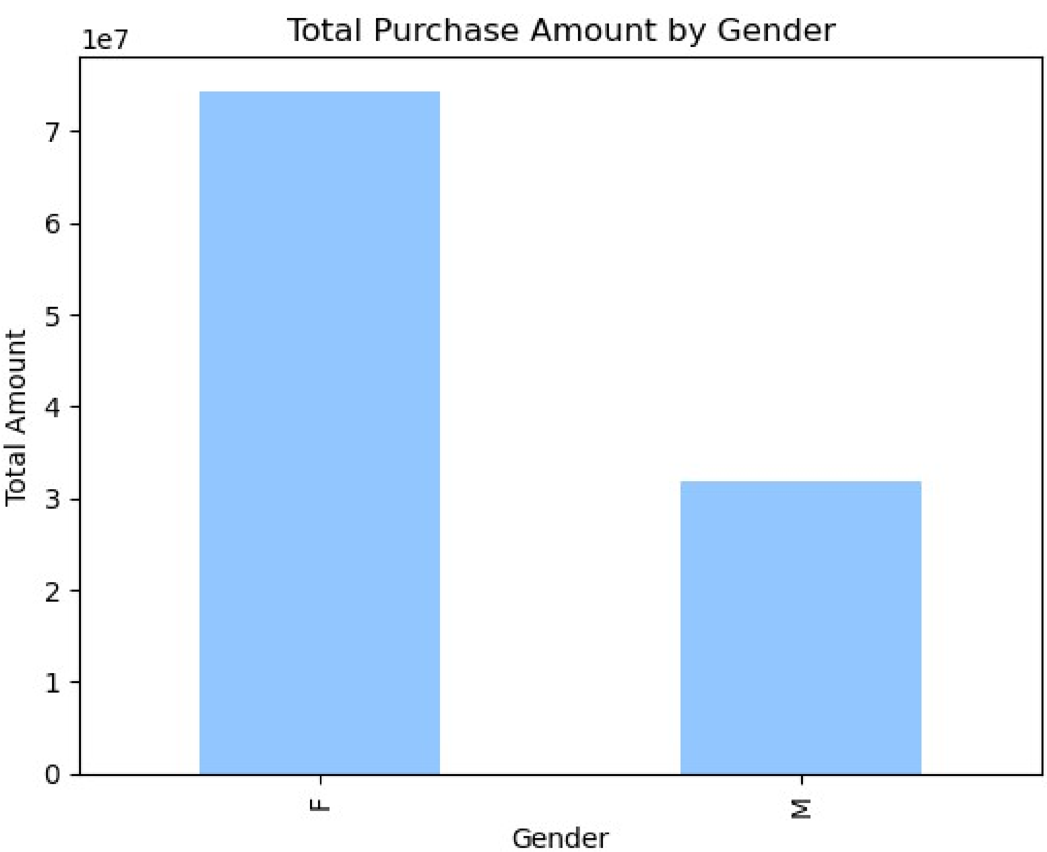
Gender Analysis

Which gender has a higher purchasing power?

```
# Count Plot
sns.countplot(data=df, x='Gender')
plt.title("Number of Orders by Gender")
plt.show()
```




```
gender_amount = df.groupby('Gender')['Amount'].sum()
gender_amount.plot(kind='bar')
plt.title("Total Purchase Amount by Gender")
plt.ylabel("Total Amount")
plt.show()
```



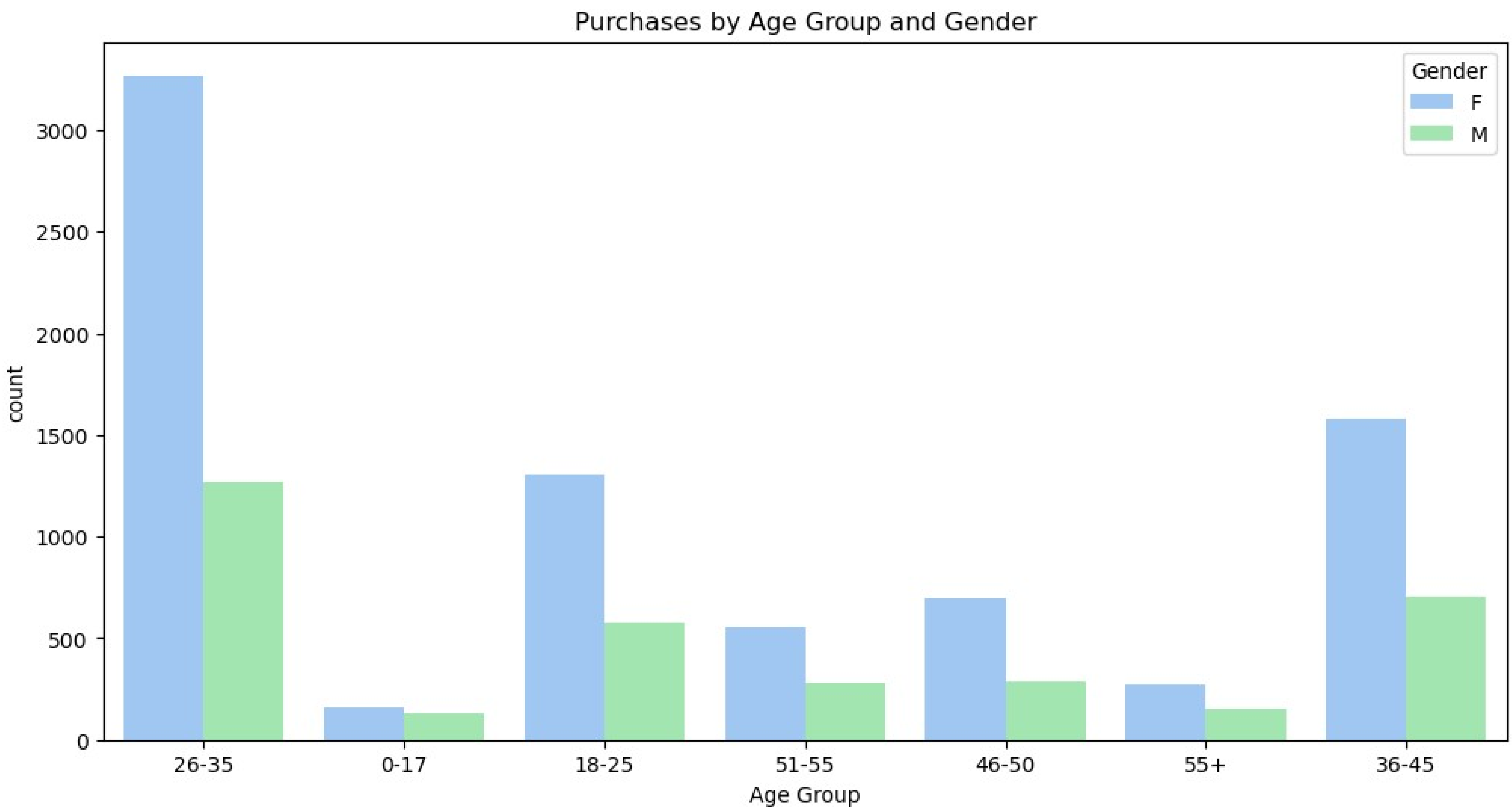
Findings Summary

- Female (F) customers generated approximately ₹7.4 crore (~74M) in total purchases
- Male (M) customers generated approximately ₹3.1 crore (~31M) in total purchases
- Females outspend males by roughly x2.4, making gender a strong differentiator in revenue contribution ~70% of total sales revenue
- Marketing and inventory decisions should prioritize female customer preferences
- Males, while likely present in reasonable order volume, have a notably lower spend per segment — worth investigating whether this is due to fewer orders or lower average order value

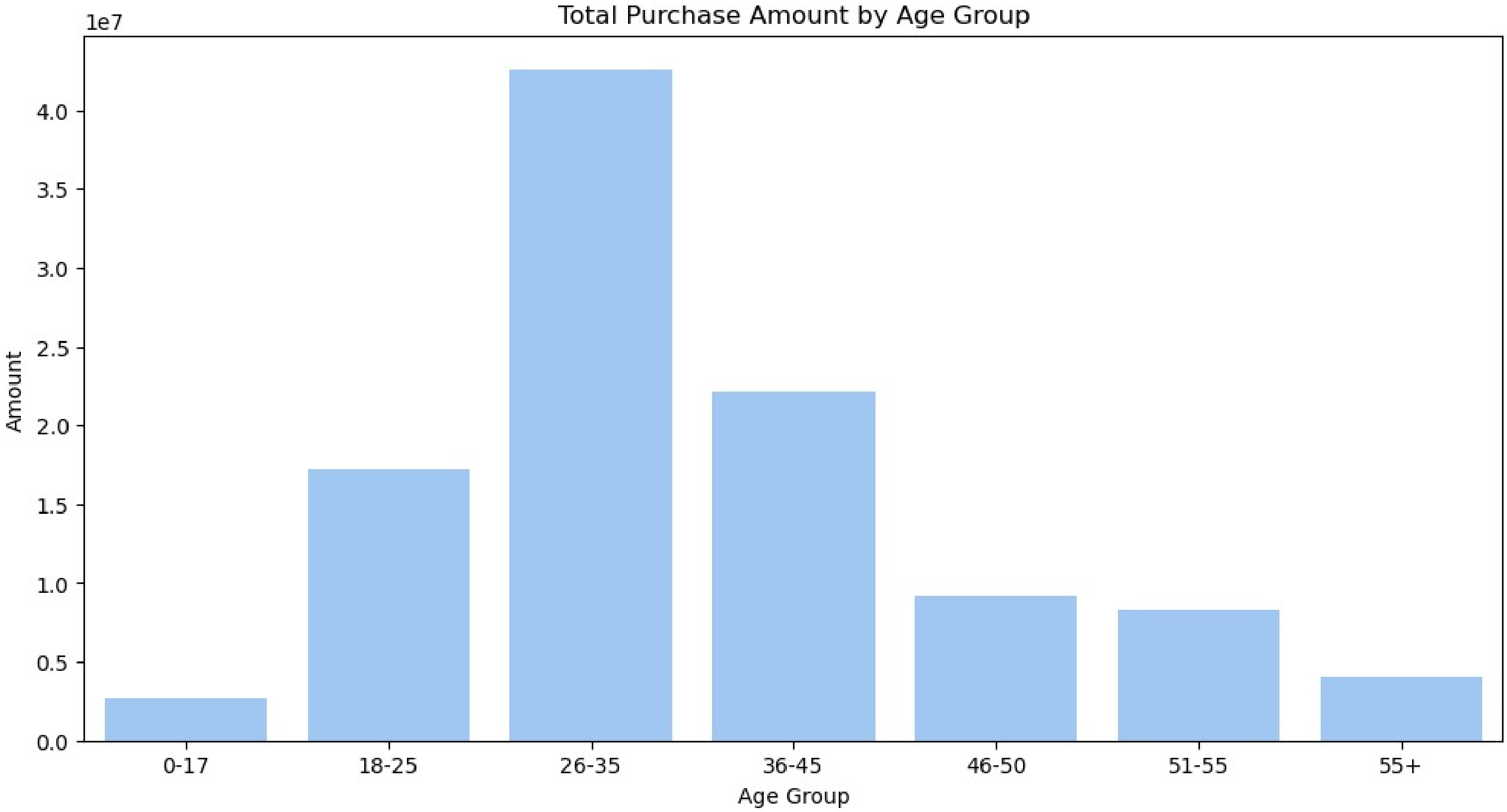
Age Group Analysis

Which age group has the most purchases, and is there a trend in purchasing power by age?

```
# count plot of age groups with hue as gender
plt.figure(figsize=(12,6))
sns.countplot(data=df, x='Age Group', hue="Gender")
plt.title("Purchases by Age Group and Gender")
plt.show()
```




```
# bar chart showing total purchase amount by age group
age_amount = df.groupby('Age Group')['Amount'].sum().reset_index()
plt.figure(figsize=(12,6))
sns.barplot(data=age_amount, x='Age Group', y='Amount')
plt.title("Total Purchase Amount by Age Group")
plt.show()
```



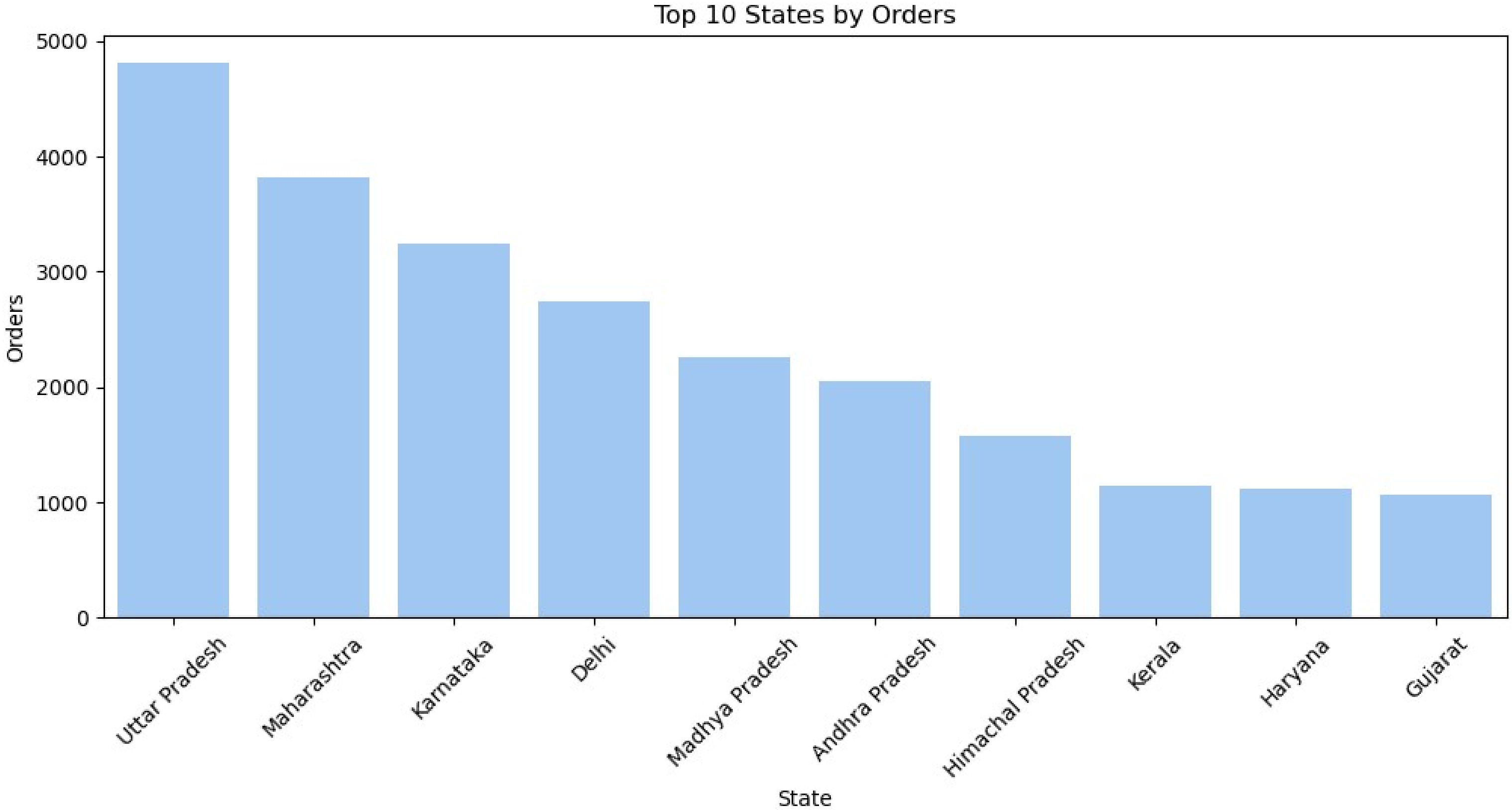
Findings Summary

- The 26–35 age group dominates, contributing nearly 40% of total purchase revenue — almost double the next closest group
- Spending drops sharply after 35, showing a clear inverted-V pattern peaking in the late 20s to mid-30s
- The 18–25 group shows decent spend, indicating younger adults are an emerging segment
- Customers 45 and older collectively contribute less than the 26–35 group alone

State Analysis

Which states generate the highest number of orders and revenue?

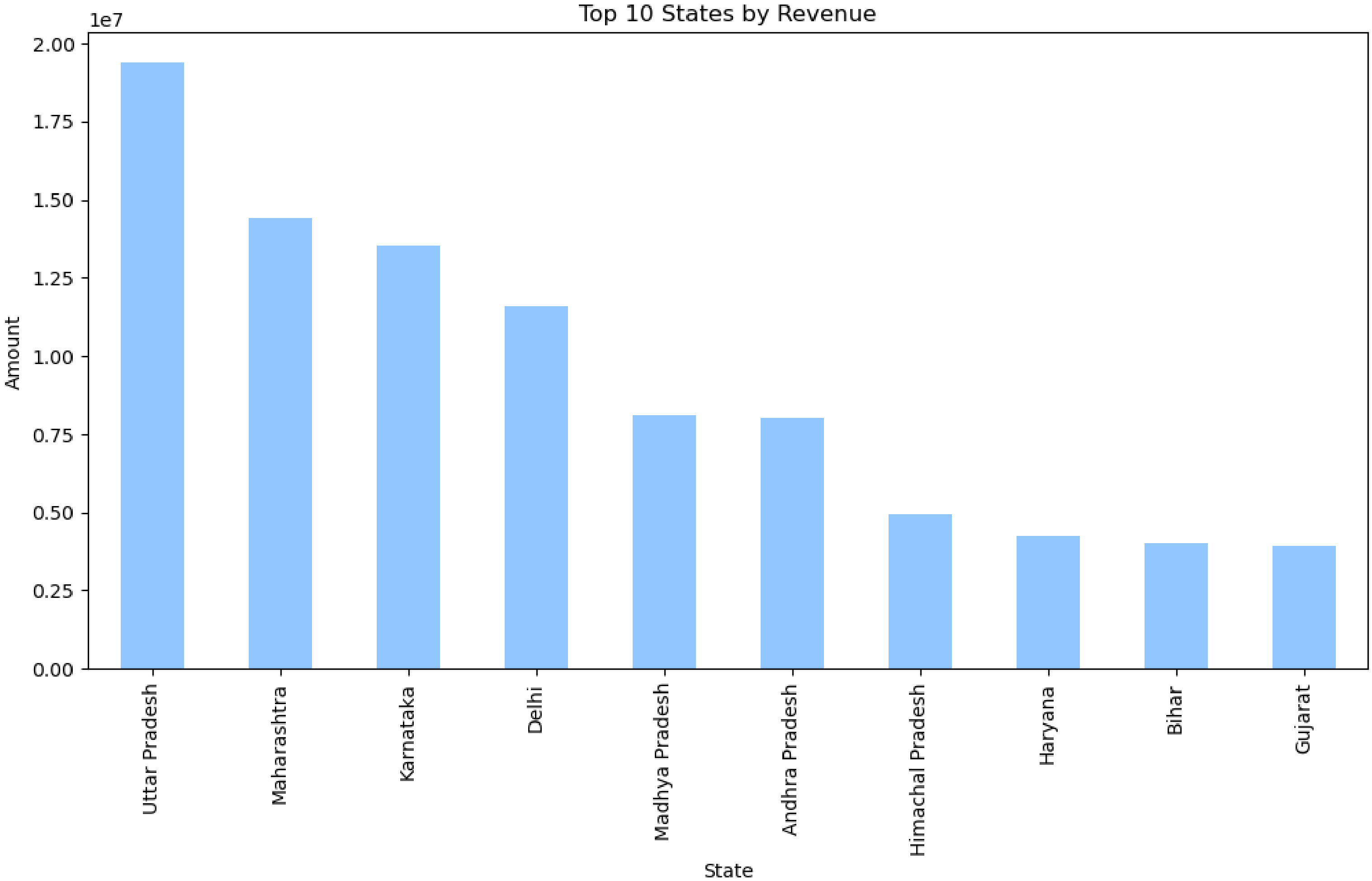
```
# bar charts for the number of orders and total amount by state
top_states_orders = df.groupby('State')['Orders'].sum().nlargest(10).reset_index()
plt.figure(figsize=(12,5))
sns.barplot(data=top_states_orders, x='State', y='Orders')
plt.xticks(rotation=45)
plt.title("Top 10 States by Orders")
plt.show()
```



Key Observations

- Uttar Pradesh leads by a wide margin, generating ~26% more orders than Maharashtra
- The top 3 states (UP, Maharashtra, Karnataka) together account for the bulk of all orders
- There's a steep drop-off after Delhi — the bottom 4 states each have fewer than 1,200 orders
- The high-order states are a mix of large population centers (UP, Maharashtra, Delhi) and tech/metro hubs (Karnataka)

```
plt.figure(figsize=(12,6))
top_state_amount = df.groupby('State')['Amount'].sum().nlargest(10)
top_state_amount.plot(kind='bar')
plt.title("Top 10 States by Revenue")
plt.ylabel('Amount')
plt.show()
```



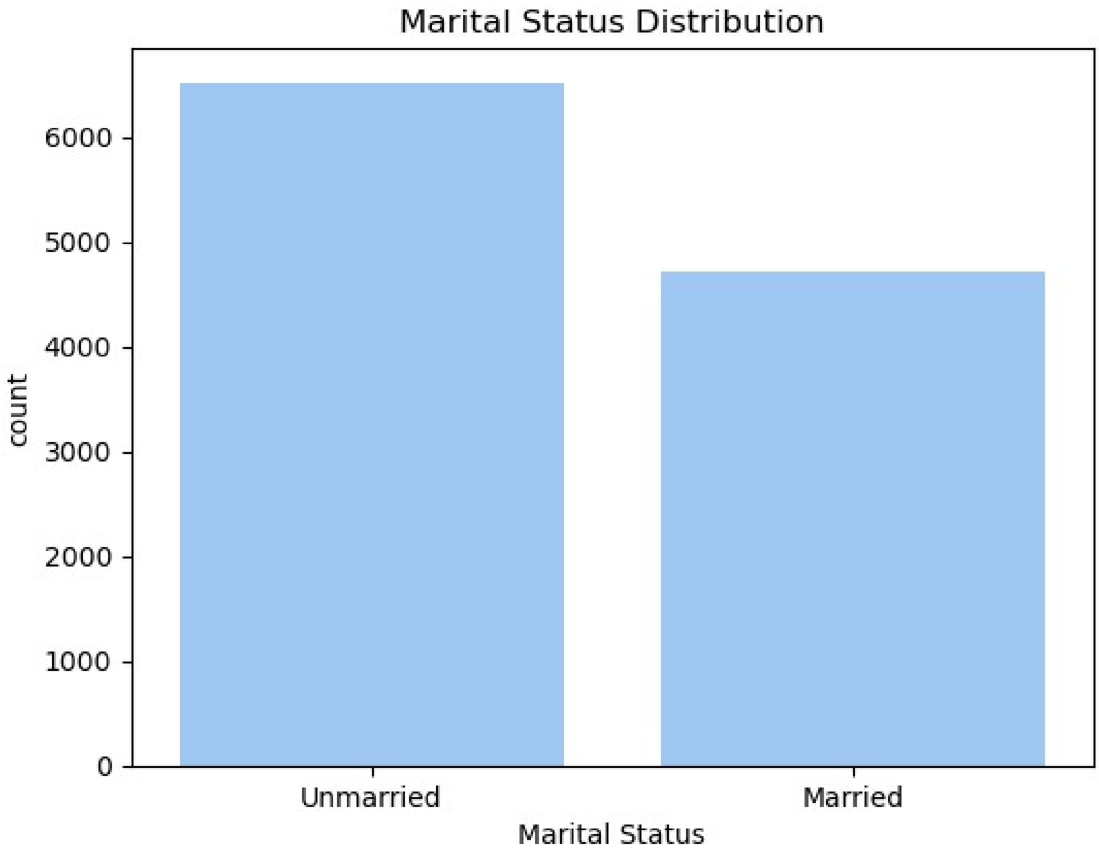
Findings Summary

- Uttar Pradesh dominates revenue too, consistent with its #1 ranking in orders — high volume is clearly translating to high revenue
- The top 4 states (UP, Maharashtra, Karnataka, Delhi) are identical across both orders and revenue charts, confirming these are the core markets
- Bihar replaces Kerala from the orders chart, suggesting Bihar customers spend more per order despite placing fewer orders overall
- Revenue drops sharply after Delhi — the bottom 6 states each generate less than ₹0.85 crore

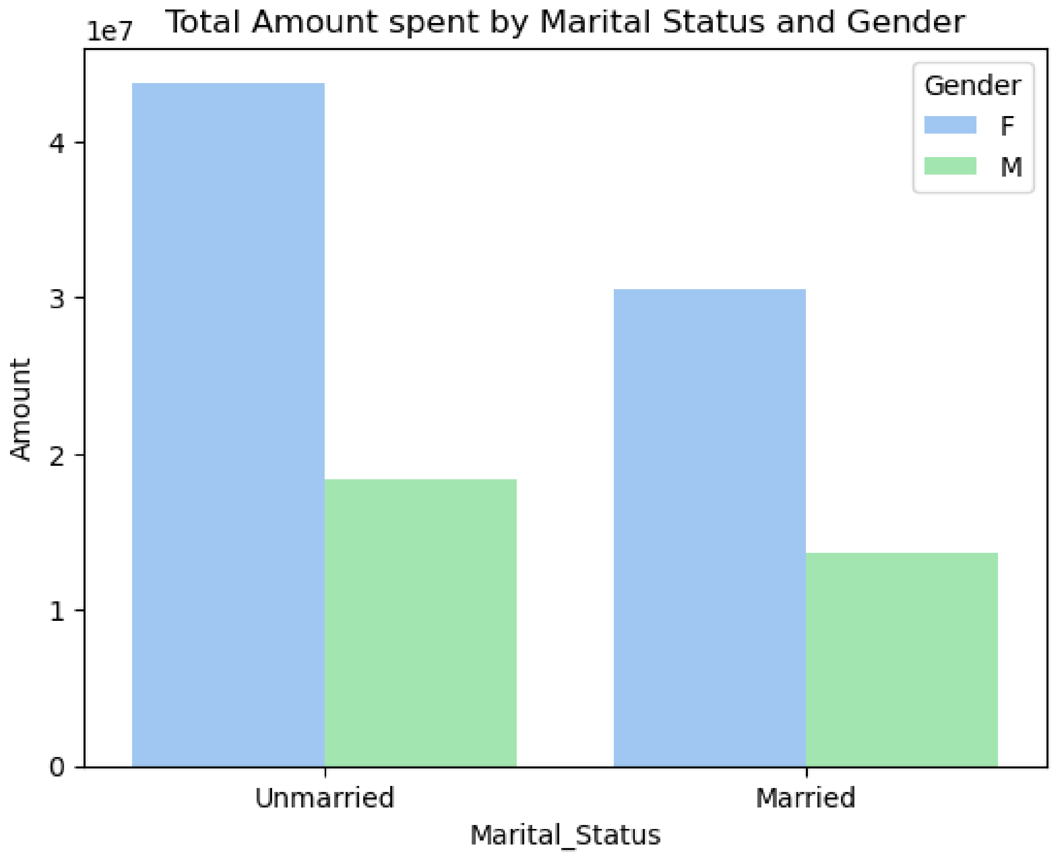
Marital Status Analysis

How does marital status affect purchasing behavior?

```
# count plot for marital status
ax = sns.countplot(data=df, x='Marital_Status')
plt.title('Marital Status Distribution')
ax.set_xticks([0,1])
ax.set_xticklabels(['Unmarried', 'Married'])
ax.set_xlabel('Marital Status')

plt.show()
```




```
# bar chart of total amount spent by marital status, with gender as a hue

marital_amount = df.groupby(['Marital_Status', 'Gender'])
['Amount'].sum().reset_index()

ax = sns.barplot(data=marital_amount, x='Marital_Status', y='Amount', hue='Gender')
plt.title("Total Amount spent by Marital Status and Gender")
ax.set_xticks([0,1])
ax.set_xticklabels(['Unmarried', 'Married'])
plt.show()
```



Finding Summary

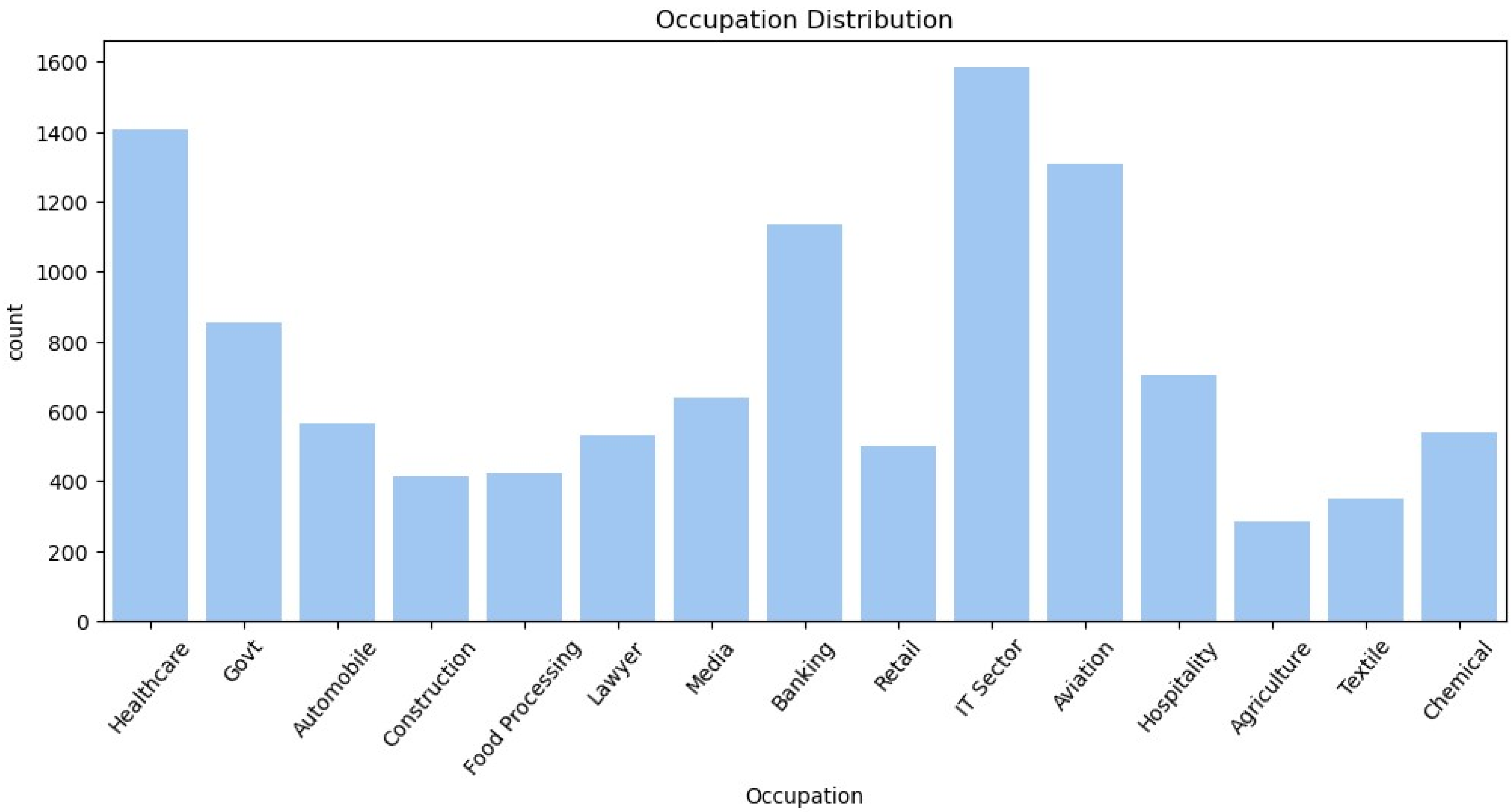
How Marital Status Affects Purchasing Behavior?

- Unmarried customers outspend married customers across both genders — unmarried females spend ~43% more than married females, and unmarried males spend ~37% more than married males
- Females dominate spending in both marital categories — unmarried females alone account for the largest single segment
- The pattern suggests unmarried individuals have higher discretionary spending, likely due to fewer shared financial obligations
- Across all four segments, the gender gap is consistent — females always outspend males regardless of marital status

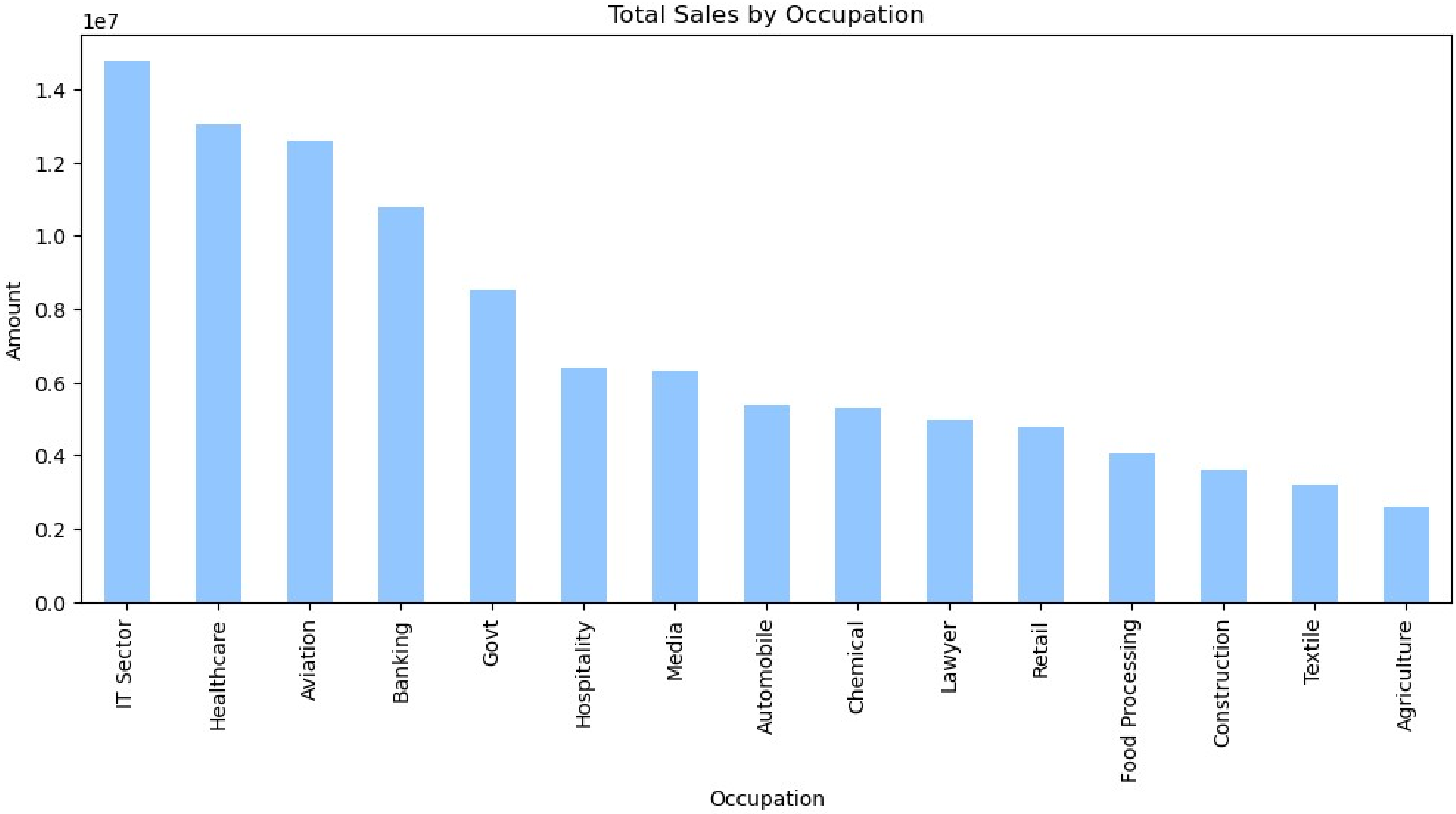
Occupation Analysis

Which occupations contribute most to sales?

```
# count plot of occupation
plt.figure(figsize=(12,5))
sns.countplot(data=df, x='Occupation')
plt.title("Occupation Distribution")
plt.xticks(rotation=50)
plt.show()
```




```
# barplot of total sales by occupation
plt.figure(figsize=(12,5))
occupation_amount = df.groupby('Occupation')
['Amount'].sum().sort_values(ascending=False)
occupation_amount.plot(kind='bar')
plt.title("Total Sales by Occupation")
plt.ylabel('Amount')
plt.show()
```



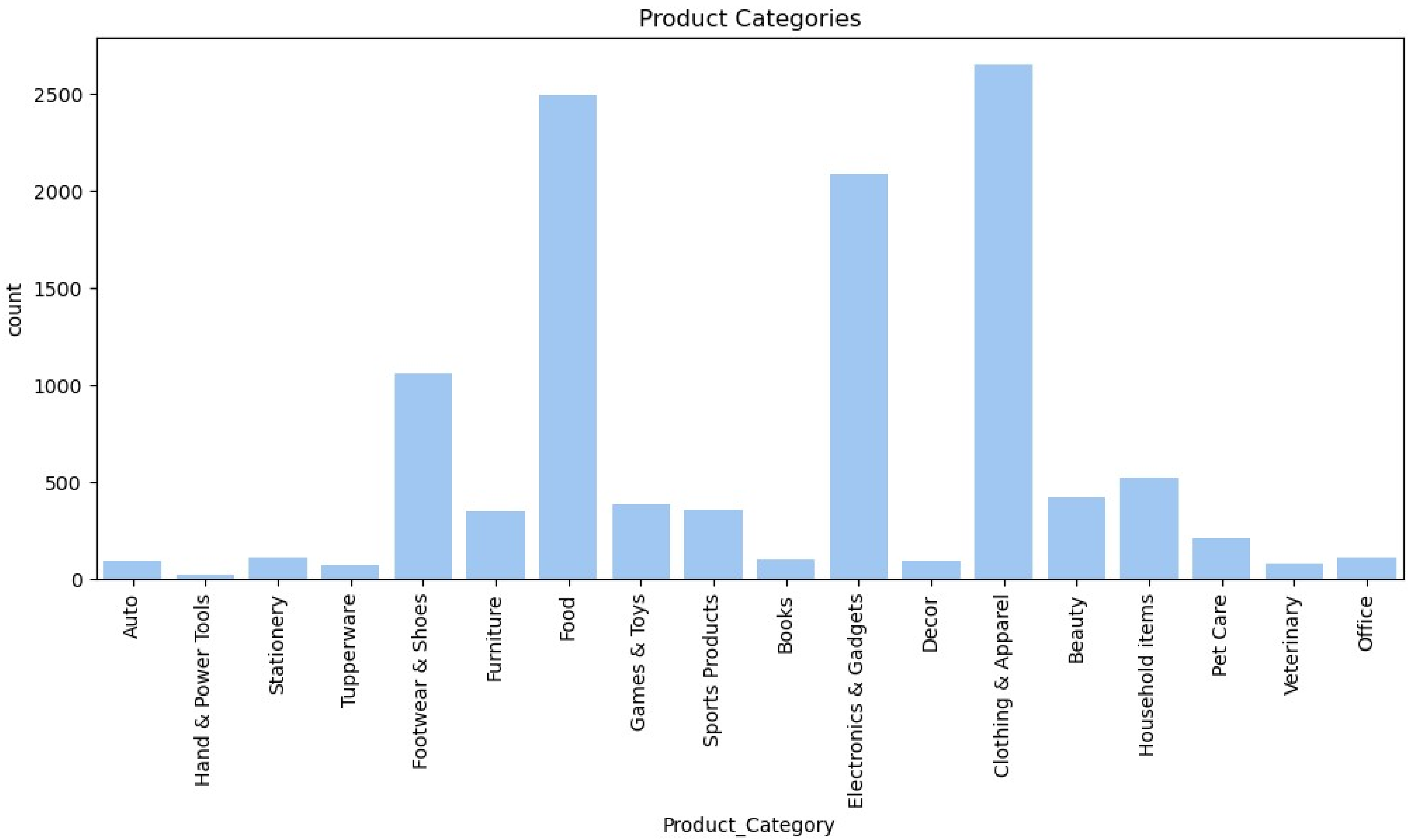
Findings Summary

- IT Sector leads all occupations, consistent with high salaries and strong online shopping habits
- The top 4 (IT, Healthcare, Aviation, Banking) are all high-income professional sectors, collectively contributing the majority of sales revenue
- There's a sharp cliff after Banking — Govt and below contribute significantly less
- Agriculture contributes the least, reflecting lower disposable income and limited digital commerce penetration

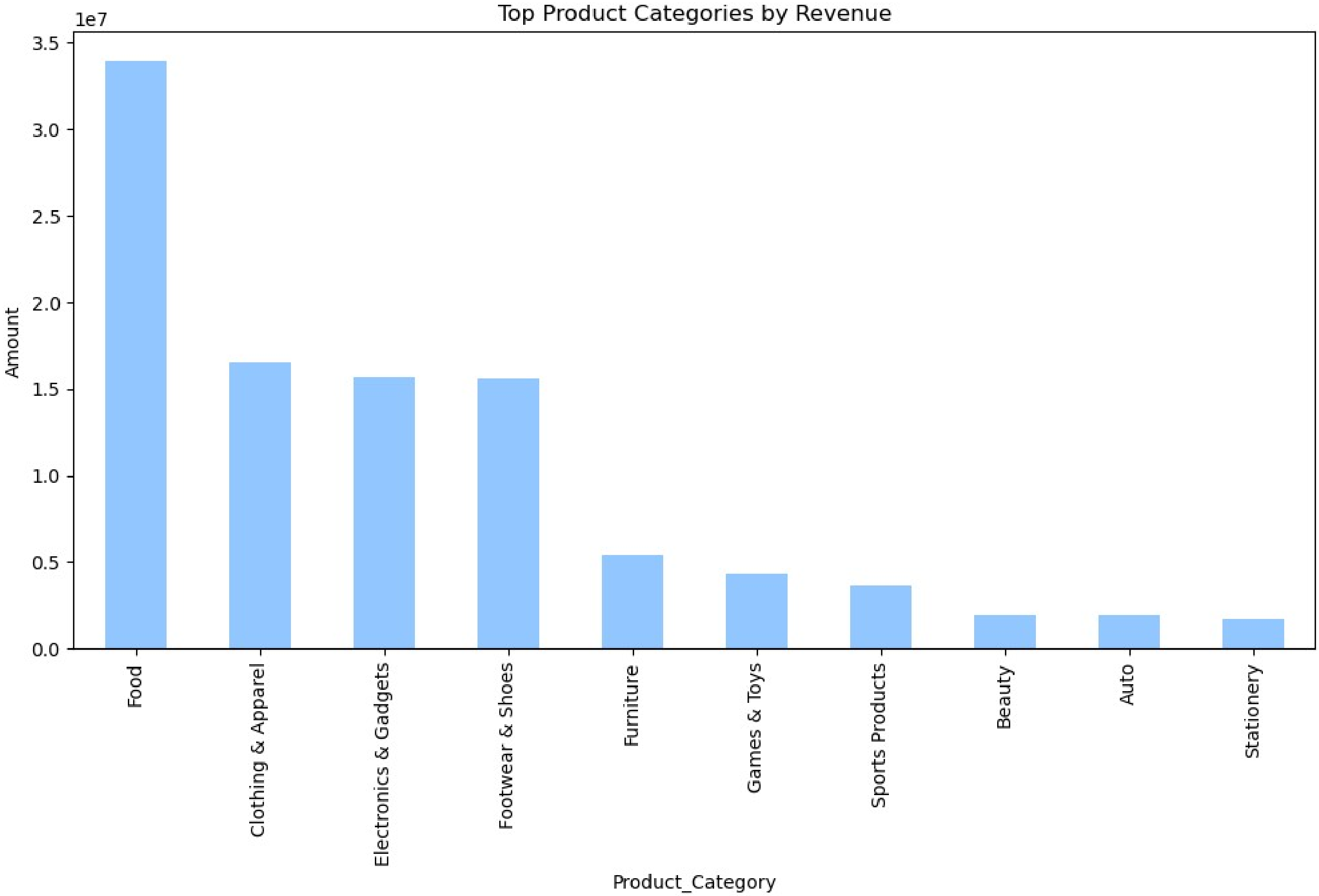
Product Category Analysis

What product categories are the most popular, and which ones generate the most revenue?

```
# count plot of product categories
plt.figure(figsize=(12,5))
sns.countplot(data=df, x='Product_Category')
plt.title('Product Categories')
plt.xticks(rotation=90)
plt.show()
```




```python
# bar chart of total amount by product category for the top categories
category_amount = df.groupby('Product_Category')['Amount'].sum().nlargest(10)
plt.figure(figsize=(12,6))
category_amount.plot(kind='bar')
plt.title("Top Product Categories by Revenue")
plt.ylabel('Amount')
plt.show()
```



Findings Summary

- Food dominates by a massive margin — generating ~2× the revenue of the next closest category, making it the single most critical product segment
- Clothing, Electronics, and Footwear form a strong second tier, each contributing ~₹1.55 crore and nearly neck-and-neck with each other
- There's a dramatic cliff after Footwear — Furniture onwards contributes less than ₹0.55 crore each
- Beauty, Auto, and Stationery are the weakest categories, each under ₹0.20 crore
- The top 4 categories together account for roughly 80% of total revenue

Conclusion

Complete High-Purchasing Customer Profile

Attribute	Profile
Gender	Female
Marital Status	Unmarried
Age Group	26–35
Occupation	IT
Sector	Healthcare, or Aviation
Location	Uttar Pradesh, Maharashtra, Karnataka, or Delhi
Top Categories	Food, Clothing & Apparel, Electronics & Gadgets, Footwear
Life Stage	Independent, salaried young professional

Attribute	Profile
Spending Driver	High disposable income; everyday essentials + lifestyle upgrades

Final Business Insight:

- Stock and promote Food, Clothing, Electronics, and Footwear heavily during New Year sales
- These four categories alone drive ~80% of revenue
- For the core segment — unmarried females aged 26–35 in IT/Healthcare — bundled offers across Clothing + Footwear + Electronics would likely yield the highest conversion and basket size

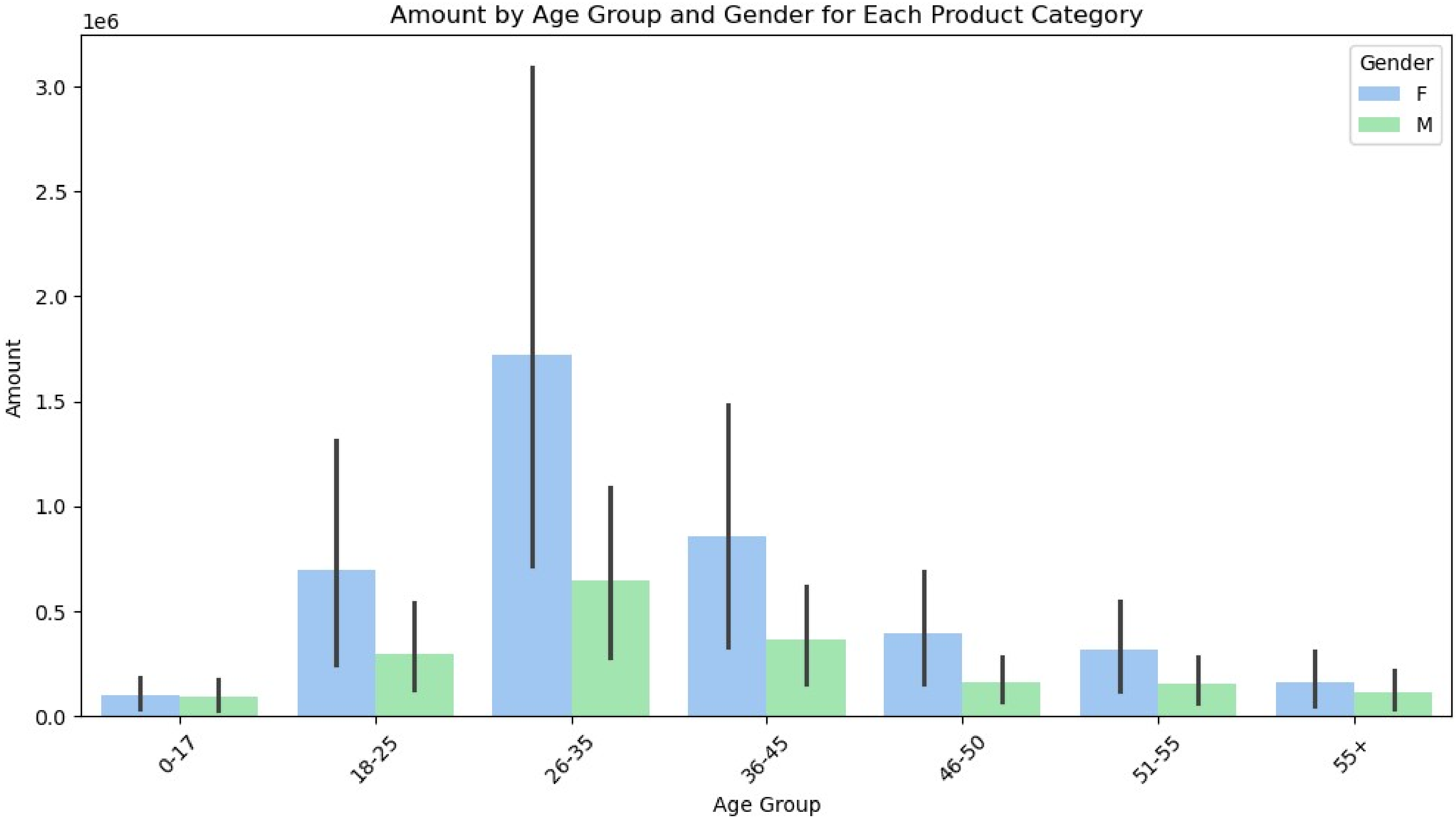
Further Exploration

Which age group contributes the most to each product category, and does this vary by gender?

```
age_cat_amount = df.groupby(['Product_Category', 'Age Group', 'Gender'])
['Amount'].sum().reset_index()
age_cat_amount.head(10)
```

	Product_Category	Age Group	Gender	Amount
0	Auto	0-17	F	9732
1	Auto	0-17	M	47151
2	Auto	18-25	F	288925
3	Auto	18-25	M	61369
4	Auto	26-35	F	541529
5	Auto	26-35	M	353578
6	Auto	36-45	F	263399
7	Auto	36-45	M	61121
8	Auto	46-50	F	61132
9	Auto	46-50	M	18608

```
plt.figure(figsize=(12,6))
sns.barplot(data=age_cat_amount, x='Age Group', y='Amount', hue='Gender')
plt.title("Amount by Age Group and Gender for Each Product Category")
plt.xticks(rotation=45)
plt.show()
```



Key Observations

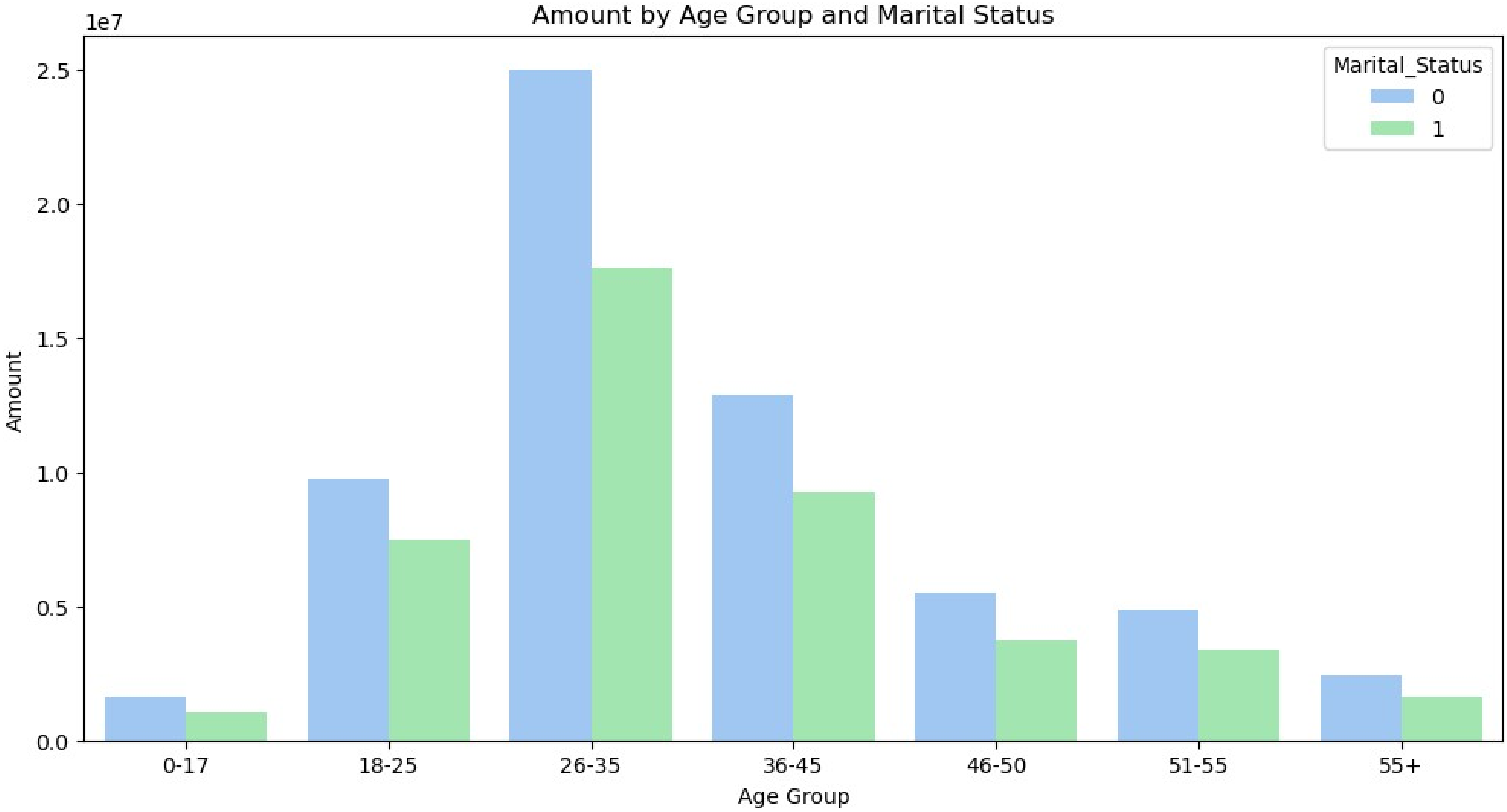
- 26–35 is the dominant age group for both genders, but female spend in this bracket (~ ₹1.72M) is nearly x3 that of males (~ ₹0.64M)
- Female dominance is consistent across every age group without exception — there is no age group where males outspend females
- The gender gap narrows in older age groups (51–55, 55+), suggesting more balanced spending behaviour post-50
- 18–25 females are a notable secondary segment, spending more than males aged 26–35

How does the amount spent vary by marital status across different age groups?

```
marital_age_amount = df.groupby(['Marital_Status', 'Age Group'])
['Amount'].sum().reset_index()
marital_age_amount.head()
```

	Marital_Status	Age Group	Amount
0	0	0-17	1655299
1	0	18-25	9781617
2	0	26-35	25008749
3	0	36-45	12870401
4	0	46-50	5472555

```
plt.figure(figsize=(12,6))
sns.barplot(data=marital_age_amount, x='Age Group', y='Amount', hue='Marital_Status')
plt.title("Amount by Age Group and Marital Status")
plt.show()
```



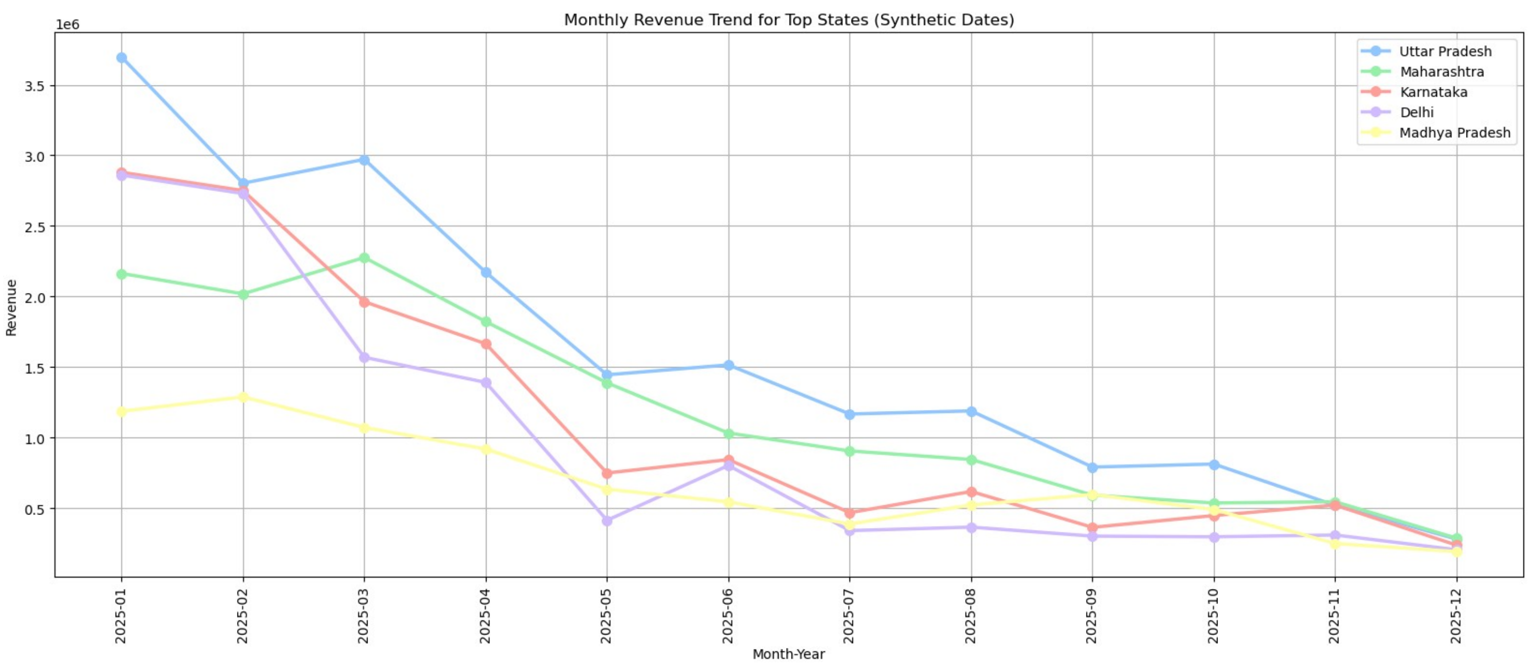
Key Observations

- Unmarried customers outspend married customers in every single age group — the pattern is universal and consistent
- The 26–35 unmarried segment is the absolute peak, spending ~₹2.50 crore — nearly 43% more than their married counterparts in the same age group
- Both marital groups follow the same inverted-V spending curve, peaking at 26–35 and declining steadily after 45
- The gap between unmarried and married narrows significantly in older age groups (46+), suggesting marital status becomes less influential as customers age
- 18–25 unmarried is a notable emerging segment — their spend (~₹0.98 crore) rivals married 36–45 customers

Which states show the highest growth in orders and revenue, and are there seasonal spikes in sales?

```
# Create a synthetic 'Order_Date' column
# This assumes orders are spread daily over a period starting from '2025-01-01'
df['Order_Date'] = pd.to_datetime(pd.date_range(start='2025-01-01', periods=len(df),
freq='D').to_series().reset_index(drop=True))

# Create Month-Year column
df['Month_Year'] = df['Order_Date'].dt.to_period('M').astype(str)

df['Month_Year'] = pd.Categorical(pd.cut(np.arange(len(df)), bins=12, labels=[f'2025-
{str(m).zfill(2)}' for m in range(1,13)]))

# Group by State and Month_Year to get orders and revenue
state_month = df.groupby(['State', 'Month_Year'], observed=True).agg(Orders=('Orders',
'sum'),Revenue=('Amount', 'sum')).reset_index()

state_month.head()
```

```
             State Month_Year  Orders  Revenue
0  Andhra Pradesh    2025-01     228  1784375
1  Andhra Pradesh    2025-02     149   968343
2  Andhra Pradesh    2025-03     117   728590
3  Andhra Pradesh    2025-04     132   639167
4  Andhra Pradesh    2025-05     244  1006059

top_state_amount = df.groupby('State')['Amount'].sum().nlargest(5).reset_index()
top_state_amount.columns = ['State', 'Revenue']

top_states = top_state_amount['State'].head(5).tolist()# from earlier

plt.figure(figsize=(16, 7))
for i, st in enumerate(top_states):
    temp = state_month[state_month['State'] == st].sort_values('Month_Year')
    plt.plot(temp['Month_Year'], temp['Revenue'],
             marker='o',
             label=st,
             linewidth=2.5,
             markersize=7)

plt.xticks(rotation=90)
plt.title("Monthly Revenue Trend for Top States (Synthetic Dates)")
plt.xlabel("Month-Year")
plt.ylabel("Revenue")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



This dataset does not contain transaction dates. Monthly revenue trends shown use synthetic dates for illustrative purposes only and do not reflect real seasonality. All other analyses (demographics, geography, product categories) are based on actual data.

Are there specific occupations that prefer particular product categories more than others?

```
occ_cat = df.groupby(['Occupation', 'Product_Category'])['Amount'].sum().reset_index()

pivot_occ_cat = occ_cat.pivot(index='Occupation', columns='Product_Category',
values='Amount').fillna(0)
pivot_occ_cat
```

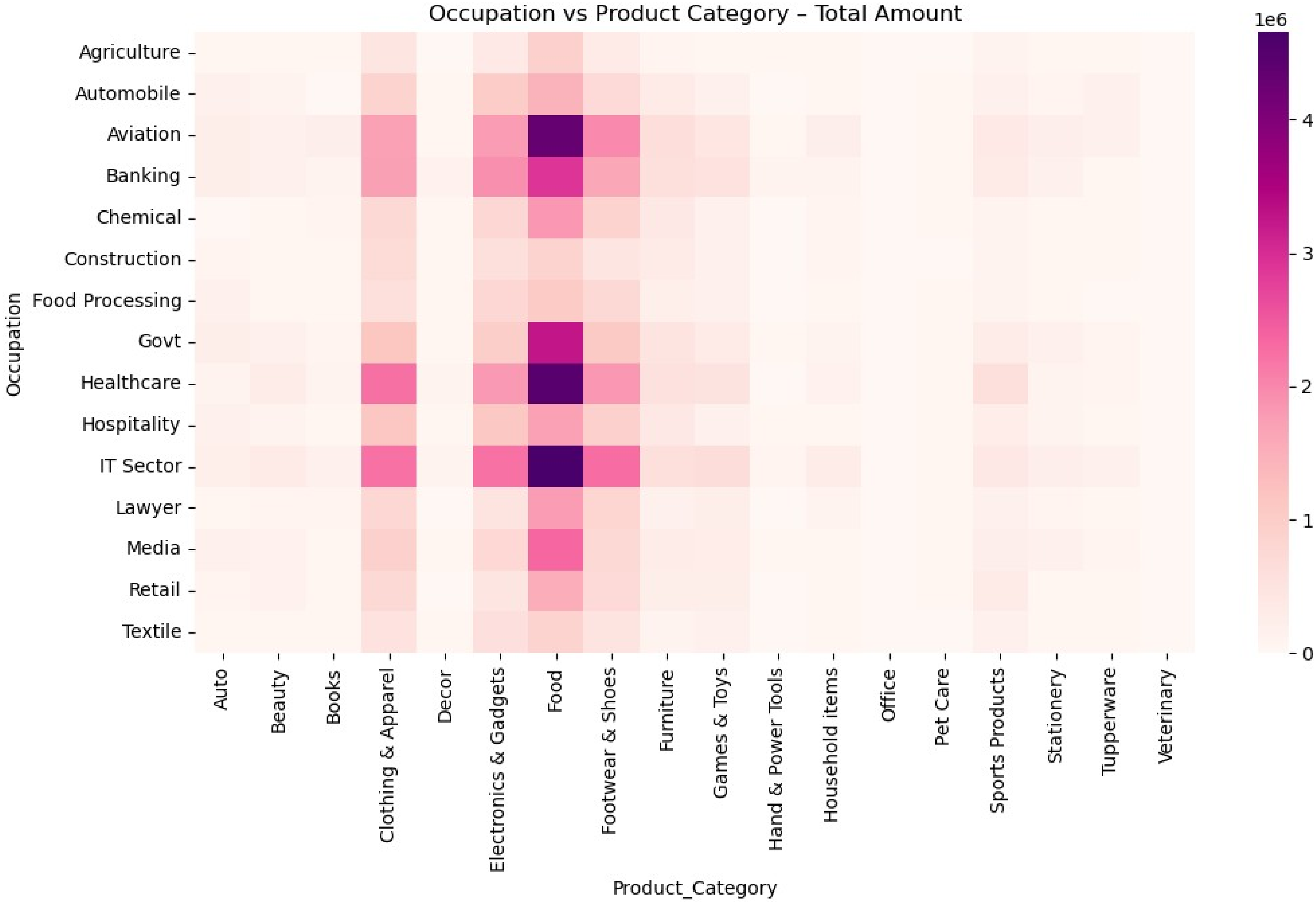
Product_Category Occupation	Auto	Beauty	Books	Clothing & Apparel	Decor
Agriculture	18609.0	43951.0	28650.0	437296.0	14057.0
Automobile	165586.0	94231.0	10364.0	870150.0	27818.0
Aviation	244129.0	184171.0	222422.0	1716254.0	59625.0
Banking	253553.0	153034.0	109905.0	1738400.0	128764.0
Chemical	0.0	53654.0	73531.0	727491.0	29494.0
Construction	80219.0	72604.0	29007.0	678794.0	36853.0
Food Processing	146326.0	45802.0	28568.0	603436.0	41789.0
Govt	245267.0	183985.0	85864.0	1158860.0	19943.0
Healthcare	103510.0	289418.0	97037.0	2249961.0	101273.0
Hospitality	150904.0	93428.0	18561.0	1131531.0	59467.0
IT Sector	211895.0	335291.0	166475.0	2226012.0	115576.0
Lawyer	71184.0	80797.0	83539.0	771556.0	6111.0
Media	163427.0	137187.0	39347.0	895328.0	41830.0
Retail	84720.0	134617.0	44908.0	744577.0	17834.0
Textile	19280.0	57314.0	23300.0	545373.0	29926.0

Product_Category Occupation	Electronics & Gadgets	Food	Footwear & Shoes
Agriculture	352614.0	930908.0	336636.0
Automobile	1006127.0	1446125.0	717888.0
Aviation	1774366.0	4306343.0	1972899.0
Banking	1919127.0	2894943.0	1611992.0
Chemical	775674.0	1821161.0	864974.0
Construction	612017.0	860256.0	461129.0
Food Processing	796316.0	1047875.0	729173.0
Govt	972346.0	3242839.0	1068102.0
Healthcare	1802020.0	4465342.0	1820931.0
Hospitality	1098596.0	1727302.0	932455.0
IT Sector	2220174.0	4655906.0	2284568.0
Lawyer	508597.0	1780666.0	835034.0
Media	750193.0	2351254.0	728628.0
Retail	457335.0	1540485.0	726085.0
Textile	598344.0	862478.0	484715.0

Product_Category Occupation	Furniture	Games & Toys	Hand & Power Tools
Agriculture	80594.0	59005.0	23160.0
Automobile	339543.0	149516.0	0.0
Aviation	618951.0	429440.0	19324.0
Banking	605507.0	536829.0	123721.0
Chemical	400051.0	195728.0	0.0
Construction	299389.0	147294.0	15151.0
Food Processing	208654.0	189455.0	0.0
Govt	481088.0	337338.0	38536.0
Healthcare	563319.0	513805.0	14627.0
Hospitality	395549.0	193569.0	33937.0
IT Sector	616630.0	646992.0	85245.0

```
Lawyer             187075.0      252884.0                   0.0
Media              277065.0      265881.0               37254.0
Retail             240251.0      249604.0               14663.0
Textile            126385.0      164354.0                   0.0

Product_Category  Household items   Office  Pet Care  Sports Products  \
Occupation
Agriculture               44079.0   4175.0   14926.0          91762.0
Automobile                72656.0    931.0   26798.0         192694.0
Aviation                 226272.0   7537.0   61840.0         360274.0
Banking                  125833.0   6415.0   34061.0         307393.0
Chemical                  76917.0   5514.0   23289.0         102553.0
Construction              74165.0   3341.0   13567.0          97884.0
Food Processing           51764.0    773.0   21384.0          99376.0
Govt                     117981.0   5499.0   36174.0         276709.0
Healthcare               141446.0  11859.0   61895.0         603478.0
Hospitality               67468.0   3413.0   30925.0         255115.0
IT Sector                287214.0  16288.0   58731.0         387421.0
Lawyer                   121994.0   5335.0   20414.0         147023.0
Media                     54370.0   6065.0   38994.0         226115.0
Retail                    52495.0   3054.0   24441.0         334112.0
Textile                   54683.0   1737.0   14838.0         154024.0

Product_Category  Stationery  Tupperware  Veterinary
Occupation
Agriculture          64976.0     42085.0      5604.0
Automobile           73226.0    162058.0     12885.0
Aviation            221390.0    166601.0     10460.0
Banking             170462.0     37405.0     13266.0
Chemical             72463.0     70683.0      4259.0
Construction         68010.0     37659.0     10172.0
Food Processing      37712.0     16311.0      5956.0
Govt                152984.0     87400.0      6297.0
Healthcare          107586.0     78600.0      8479.0
Hospitality         102354.0     71217.0     10614.0
IT Sector           226817.0    196154.0     17690.0
Lawyer               81314.0     25338.0      2804.0
Media               193386.0     88770.0       738.0
Retail               64829.0     45682.0      3478.0
Textile              38542.0     29679.0         0.0
```

```python
plt.figure(figsize=(12,6))
sns.heatmap(pivot_occ_cat, annot=False, cmap= 'RdPu')
plt.title("Occupation vs Product Category — Total Amount")
plt.show()
```



Key Observations

Food dominates across all occupations

- The entire Food column is noticeably darker than any other — it's the universal top category regardless of profession
- IT, Healthcare, and Aviation show the darkest Food cells, consistent with their higher overall spend

Electronics & Gadgets is occupation-specific

- Strongest in IT Sector, Healthcare, and Aviation — high-income, tech-comfortable professions
- Nearly invisible for Agriculture, Textile, Construction — lower digital affinity segments

Clothing & Apparel shows broad but moderate appeal

- Visible across most occupations with light-to-medium shading
- No single occupation dominates this category

The right half of the heatmap is almost entirely blank

- Categories like Books, Decor, Office, Pet Care, Tupperware, Veterinary generate near-zero spend across all occupations
- These are either niche or not relevant to New Year shopping behavior

Agriculture, Textile, Construction are uniformly light

- Consistent with their low overall revenue contribution
- No strong category preference visible — low spend spread thinly

Strategic Insight:

- Since Food is universally dominant, it should anchor all campaigns regardless of occupation targeting
- For IT/Healthcare/Aviation specifically, bundling Food + Electronics + Footwear offers would capture their full spending pattern
- Lower-income occupations (Agriculture, Textile) need value-focused food promotions rather than lifestyle or tech upsells.

What is the correlation between age and spending amount, and does this differ by gender?

```
df[['Age', 'Amount']].corr()
```

	Age	Amount
Age	1.000000	0.030941
Amount	0.030941	1.000000

```
plt.figure(figsize=(12,6))
sns.scatterplot(data=df, x='Age', y='Amount', hue='Gender', alpha=0.5)
plt.title("Age vs Amount by Gender")
plt.show()
```

Key Observations

- Correlation = 0.031 (Age alone cannot predict how much someone spends)
- Uniform spend across ages (All age groups have both high and low spenders)
- No gender-age interaction (Gender and age don't combine to create spending trends)
- Dense 15–50 cluster (This is where volume comes from, not higher per-order spend)

Key Insight:

- Spending amount in this dataset is driven by what people buy (product category) and who they are (occupation, marital status) — not how old they are
- Age groups differ in total revenue only because of differences in customer count, not because older or younger customers spend more per transaction.